

Graphical Models

Project Instructions

- **Report Submission:**
 - **Due on Sunday 7/7, 11:59PM**
 - **Submit to: pgms2024@163.com**
 - **Subject line:** UCAS Summer 2024: Graphical Models Project by [Name (ID)]
- **Project Report:**
 - The report should be prepared by either Latex (recommended) or MS Word
 - The report should be in the format of a short paper. The main text (not including references and appendix) should be **5 ~ 10 pages with 12pt font and single spacing**.
 - The report may be written in either English or Chinese.
 - The report should include the following parts: *Title; Author/Affiliation/Contact information; Abstract; Keywords; Introduction; Methods; Results; Discussion; References; Appendices* (secondary figures, tables, detailed derivations, etc.)
 - **R or Python** scripts should be submitted as a separate file with necessary documentation. The scripts should be self-explanatory, self-contained and ready to run
- **Project Options:**
 - **Option A: Literature review and simulation study:**
 - Conduct a literature review on a topic of graphical models such as high dimensional Gaussian graphical model learning, DAG/Bayesian network learning, graphical models with mixed types of nodes, time-varying graphical models, etc. Conduct a simulation study to investigate and compare (some of) the relevant methods. You may use existing packages/implementations
 - The report should include: A brief literature review; A brief description of each method under investigation; Design of the simulation study; Summary of results and findings; Conclusions: pros and cons of each method (accuracy, precision/recall, computational cost, practical aspects such as the ease of tuning, relative performance, etc.). Are the results consistent with existing literature or are there new findings? Comments and thoughts for possible improvements or extensions
 - **Option B: Methods implementation and benchmarking:**
 - Implement an algorithm in graphical models, e.g., the *hill climbing* algorithm for DAG structural learning
 - Compare with an existing implementation, e.g., the implementation from the *bnlearn* R package
 - Illustrate advantages (e.g., faster, more scalable, more robust, more features, etc.) of your implementation through a numerical study

- The report should include: An overview of the algorithm; Implementation details and key improvements; Benchmarking results and conclusions; Comments and thoughts for possible improvements or extensions
- **Some useful R packages:**
 - R-package for graph visualization: *igraph*
 - R package for DAG learning: *bnlearn*
 - R packages for high-dimensional Gaussian graphical models (GGM): *glasso* (penalized likelihood), *clime* (Constrained L1-Minimization for Inverse (Covariance) Matrix Estimation), *space* (penalized pseudo likelihood), etc.